UGH/INT/31/07/26

RollNo:.....

MARAUGUSTHINOSECOLLEGERAMAPURAM

MGU-BCA(H) INTERNAL EXAMINATION,JULY 2026

MG3CCRBCA203–DESIGN AND ANALYSIS OF ALGORITHMS

Third Semester

Time: 1 Hour **MaximumMarks:35**

(Remember (K), Understand (U), Apply (A), Analyse (An))

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

PartA

(Answer all questions. Each Question carries 2 Marks).

1. Define algorithm. [K][1]
2. Explain Recursion. [A][1]
3. Explain Big Oh Notation. [A][1]
4. Explain best, worst and average case complexity. [A][1]
5. What is the complexity of binary search algorithm? [U][2]

(5 X 2=10)

PART B

(Answer any THREE questions. Each question carries 5 marks)

6. Explain Different Algorithm Design Techniques. [A][1]
7. Explain Space And Time Complexity of an Algorithm. [A][1]
8. Explain Asymptotic Notation. [A][1]
9. Discuss Control abstraction for Divide and Conquer Method. [An][2]

(3X 5=15)

PART C

(Answer any One question. Each question carries 10 marks)

10. Explain Merge Sort Algorithm with an Example. [A][2]
11. Explain the algorithm for finding maximum and minimum in an array with Example. [A][2]

(1 X 10= 10)

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QP CODE: MOD12102026

RegNo:.....

Name:.....

MAR AUGUSTHINOSE COLLEGE, RAMAPURAM

MGU-BCA(H) MODEL EXAMINATION, October 2026

MG3CCRBCA203–DESIGN AND ANALYSIS OF ALGORITHMS

Third Semester

Maximum Marks : 50 **Time : 1.5 Hours**

(Remember(K),Understand(U),Apply(A),Analyse(An))

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

PartA

Answer All questions. Each question carries 2 marks.

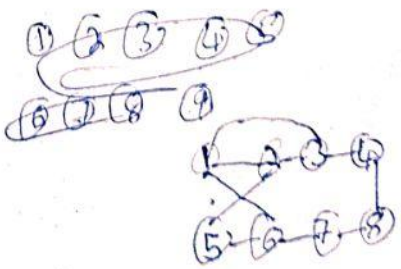
1.	Define control abstraction?	[k][1]
2.	Define implicit and explicit constraints.	[k][3]
3.	Differentiate between greedy and dynamic programming?.	[u][1]
4.	Explain the control abstraction of backtracking.	[u][1]
5.	Define NQueen's problem.	[k][1]
6.	Define optimal solution and feasible solution.	[k][2]
7.	Define Principle of Optimality.	[k][3]
8.	Define Spanning tree	[An][2,3]
9.	Write the objective function and feasibility constraints of knapsack problem.	[u][2]
10.	Define Live node, E-node & Dead node.	[k][3]

(10x2=20)

PartB

Answer any THREE questions.Each question carries 5 marks.

11.	Write the algorithm for all-pairs shortest path.	[u][3]
12.	ExplainTime complexity and Space Complexity	[u][3]
13.	Writean algorithm for Knapsack Problem	[u][3]
14.	Write an algorithm for binary search.	[u][2]
15.	Explain algorithm for DFS and BFS.	[u][3]
16.	Explain Hamiltonian cycle.	[u][3]



(3x5=15)

PartC

AnsweranyONEquestions.Eachquestioncarries15marks.

17.	Explain Sum of Subset Problem with example.	[u][2]
18.	Explain Prim's algorithm for finding MST.?	[u][2]

(1x15=15)